North South University

Boolean Algebra &

Logic Gates

Digital Logic Design | CSC 231

Summer 2026 • Week 2.2

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Section 1

Boolean Algebra

What is Boolean Algebra?

Definition

- A closed algebraic system with set K = {0,1} and two operators: · (AND) and + (OR)
- Developed by George Boole (1854); applied to logic circuits by Claude Shannon (1938)

Why it matters for CSC 231

- Every digital circuit — from a simple gate to a CPU — is described by Boolean algebra
- Allows us to analyze, simplify, and implement switching functions as hardware

Two logic values only

- Logic 0 → low voltage (~0V); Logic 1 → high voltage (~3.3V or 5V)
- Binary signals simplify circuit implementation enormously

Variables & Literals

- Boolean variable: holds value 0 or 1 (e.g. A, B, x, y)
- Literal: a variable in true (A) or complemented (A') form



wo binary operators, + and •, (Huntington) postulates:
. (a) The structure is closed with respect to the operator +.

George Boole (1854) · E.V. Huntington (1904) · C.E. Shannon (1938)

Postulates of Boolean Algebra

Postulate 1 — Closure

- If A, B ∈ {0,1} then A+B ∈ {0,1} and A·B ∈ {0,1}

Postulate 2 — Identity Elements

- A + 0 = A (identity for OR)
- A · 1 = A (identity for AND)

Postulate 3 — Commutativity

- A + B = B + A A · B = B · A

Postulate 4 — Associativity

- A+(B+C) = (A+B)+C A·(B·C) = (A·B)·C

Postulate 5 — Distributivity

- A·(B+C) = AB+AC A+(B·C) = (A+B)·(A+C)

Postulate 6 — Complement

- A + A' = 1 A · A' = 0

Fundamental Theorems (Single Variable)

Theorem 1 — Idempotency

- $A + A = A$ $A \cdot A = A$

Theorem 2 — Null Element

- $A + 1 = 1$ $A \cdot 0 = 0$

Theorem 3 — Involution (Double Complement)

- $(A')' = A$

Properties of 0 and 1

- $0 + A = A$ $1 \cdot A = A$
- $1 + A = 1$ $0 \cdot A = 0$
- $0' = 1$ $1' = 0$

Complement

- $A + A' = 1$ $A \cdot A' = 0$

Theorems — Multi-Variable (Part 1)

Theorem 4 — Absorption

- A + AB = A A(A+B) = A
  - Example: (X+Y) + (X+Y)Z = X+Y
  - Example: AB'(AB'+B'C) = AB'

Theorem 5 — Simplification

- A + A'B = A + B A(A'+B) = AB
  - Example: B + AB'C'D = B + AC'D

Theorem 6 — Adjacency

- AB + AB' = A (A+B)(A+B') = A
  - Example: ABC + AB'C = AC

Theorem 7

- AB + AB'C = AB + AC (A+B)(A+B'+C) = (A+B)(A+C)
  - Example: wy' + wx'y + wxy + wxz' = w + wxz' = w

De Morgan's Theorem

First Theorem

- (A + B)' = A' · B' — NOT of a sum = product of NOTs

Second Theorem

- (A · B)' = A' + B' — NOT of a product = sum of NOTs

Generalized De Morgan

- (A+B+…+Z)' = A'·B'·…·Z'
- (A·B·…·Z)' = A'+B'+…+Z'

Extended Example

- (a + bc)' = a'(bc)' = a'(b'+c') = a'b' + a'c'
- (a(b+c)+a'b)' = (ab+ac+a'b)' = (b+ac)' = b'(a'+c')

Verification via Truth Table

- x y (x+y)' x'y' (xy)' x'+y'
  - Columns (x+y)' and x'y' are identical — theorem confirmed

Theorem 8 — Consensus Theorem

Statement

- AB + A'C + BC = AB + A'C (BC is the consensus term — redundant)
- (A+B)(A'+C)(B+C) = (A+B)(A'+C)

Proof (SOP form)

- BC = BC·1 = BC(A+A') = ABC + A'BC
- AB + A'C + ABC + A'BC
  - = AB(1+C) + A'C(1+B) = AB + A'C ✓

Examples from Chapter 2

- AB + A'CD + BCD = AB + A'CD
- (a+b')(a'+c)(b'+c) = (a+b')(a'+c)
- ABC + A'D + B'D + CD = ABC + A'D + B'D

Application

- Identifies and removes redundant terms in SOP/POS expressions
- Essential tool for algebraic minimization

Duality Principle & Summary Table

Duality Principle

- Swap AND ↔ OR and 0 ↔ 1 in any identity → another valid identity

Examples

- A+0=A ↔ A·1=A
- A+1=1 ↔ A·0=0
- A+A'=1 ↔ A·A'=0
- (A+B)'=A'B' ↔ (AB)'=A'+B'

Why useful?

- Every Boolean law comes in a dual pair — learn one, get the other free

Quick Reference — Boolean Identities

- Identity: A+0=A, A·1=A
- Null: A+1=1, A·0=0
- Idempotent: A+A=A, A·A=A
- Complement: A+A'=1, A·A'=0
- Involution: (A')'=A
- Commutative: A+B=B+A
- Associative: (A+B)+C=A+(B+C)
- Distributive: A(B+C)=AB+AC
- Absorption: A+AB=A
- De Morgan: (A+B)'=A'B'
- Consensus: AB+A'C+BC=AB+A'C

Section 2

Boolean Functions & Truth Tables

Boolean Functions — Definition & Forms

Definition

- A Boolean function maps n binary input variables to a binary output: F:{0,1}^{n}→{0,1}
- For n variables: 2^n possible input combinations; 2^(2^n) distinct functions

SOP — Sum of Products

- OR of product (AND) terms
  - Example: F1 = x + yz' (F1=1 if x=1, or y=0 and z=1)

POS — Product of Sums

- AND of sum (OR) terms
  - Example: F2 = x'y'z + x'yz + xy' (simplified: F2 = x'z + xy')

Representations

- 1. Algebraic expression 2. Truth table
- 3. Logic diagram (circuit) 4. Canonical (minterm/maxterm) form

- The binary combinations for th
by counting from 0 through 2^n -

Table 2.2
Truth Tables for F_1 and F_2

x	y	z	F_1	F_2
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	0
1	1	1	1	0

Truth Table Construction — Example

x	y	z	y'
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

- Distributive laws:
  - $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$
  - $x + (y \cdot z) = (x+y) \cdot (x+z)$

x	y	z
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

y + z	x · (y + z)
0	0
1	0
1	0
1	0
0	0
1	1
1	1
1	1

x · y	x · z	(x
0	0	
0	0	
0	0	
0	0	
0	0	
0	1	
1	0	
1	1	

y · z	x+(y · z)
0	0
0	0
0	0
1	1
0	1
0	1
0	1
1	1

x+y	x+z	(x
0	0	
0	1	
1	0	
1	1	
1	1	
1	1	
1	1	
1	1	

Truth Table Construction — Example

- Distributive laws:
  - $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$
  - $x + (y \cdot z) = (x+y) \cdot (x+z)$

x	y	z
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$y + z$	$x \cdot (y + z)$
0	0
1	0
1	0
1	0
0	0
1	1
1	1
1	1

$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	1	1
1	0	1
1	1	1

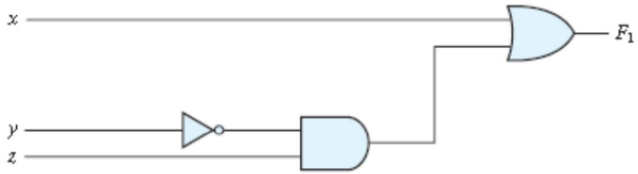
$y \cdot z$	$x+(y \cdot z)$
0	0
0	0
0	0
1	1
0	1
0	1
0	1
1	1

$x+y$	$x+z$	$(x+y) \cdot (x+z)$
0	0	0
0	1	0
1	0	0
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1

Truth Tables for F1 = x+yz' and F2 = x'y'z+x'yz+xy' (Table 2.2)

x	y	z	F1	F2		
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	0	0	0	0
0	1	1	0	1	0	1
1	0	0	0	0	1	1
1	0	1	0	0	0	0
1	1	0	0	0	1	1
1	1	1	1	0	0	1

Implementation of F_1 with logic gates



$F_1 = x + y z'$

Fig. 1.1 — Implementation of $F_1 = x + yz'$

Complementing Boolean Functions

Method 1 — Apply De Morgan's Theorem

- Complement all literals AND swap + ↔ ·
  - Example: F = xy'+x'z
  - F' = (xy')' · (x'z)' = (x+y)(x+z')

Method 2 — Dual + Complement Literals

- Step 1: Write the dual of F (swap +↔· and 0↔1 in constants)
- Step 2: Complement every literal
  - Example: F = AB + A'C
  - Dual: (A+B)(A'+C)
  - Complement literals: (A'+B')(A+C') = F' ✓

Switching Function Complement (from Ch.2)

- For f(A,B,Q,Z) = Σm(0,1,6,7)
  - f'(A,B,Q,Z) = Σm(2,3,4,5,8,9,10,11,12,13,14,15)

EXAMPLE 2.2

Find the complement of the functions $F_1 = x'yz' + x'y'z$ and $F_2 = x(y'z' + yz)$. By applying DeMorgan's theorems as many times as necessary, the complements are obtained as follows:

$$F_1' = (x'yz' + x'y'z)' = (x'yz')'(x'y'z)' = (x + y' + z)(x + y + z')$$
$$F_2' = [x(y'z' + yz)]' = x' + (y'z' + yz)' = x' + (y'z')'(yz)'$$
$$= x' + (y + z)(y' + z')$$
$$= x' + yz' + y'z$$

EXAMPLE 2.3

Find the complement of the functions F_1 and F_2 of Example 2.2 by taking their duals and complementing each literal.

1. $F_1 = x'yz' + x'y'z$.
   The dual of F_1 is $(x' + y + z')(x' + y' + z)$.
   Complement each literal: $(x + y' + z)(x + y + z') = F_1'$.
2. $F_2 = x(y'z' + yz)$.
   The dual of F_2 is $x + (y' + z')(y + z)$.
   Complement each literal: $x' + (y + z)(y' + z') = F_2'$.

Section 3

Logic Gates & Symbols

AND, OR, and NOT Gates

AND Gate

- Output = 1 only when ALL inputs are 1
- Expression: F = A·B (or simply AB)
- Truth table: 00→0, 01→0, 10→0, 11→1

OR Gate

- Output = 1 when AT LEAST ONE input is 1
- Expression: F = A + B
- Truth table: 00→0, 01→1, 10→1, 11→1

NOT Gate (Inverter)

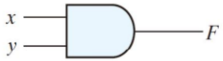
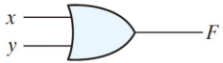
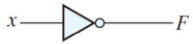
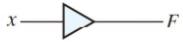
- Output = complement of input
- Expression: F = A' Truth table: 0→1, 1→0

Physical Realisation

- Early computers used relay switches; today CMOS transistors implement all gates
- CMOS: Complementary Metal-Oxide Semiconductor — fast, low power

AND, OR, and NOT Gates

Digital Logic Gates of Two Inputs

Name	Graphic symbol	Algebraic function	Truth table
AND		$F = x \cdot y$	x y | F 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1
OR		$F = x + y$	x y | F 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 1
Inverter		$F = x'$	x | F 0 | 1 1 | 0
Buffer		$F = x$	x | F 0 | 0 1 | 1

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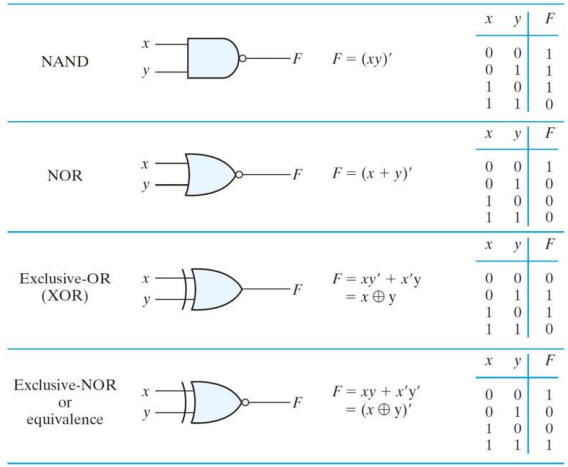
All Seven Gate Truth Tables

A	B	AND AB	OR A+B	NAND (AB)'	NOR (A+B)'	XOR A⊕B	XNOR (A⊕B)'
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Digital Logic Gates of Two Inputs



NAND and NOR — Universal Gates

NAND as Universal Gate

- NOT A: (A·A)' = A'
- AND AB: ((AB)')' = AB
- OR A+B: (A'·B')' = A+B [De Morgan]

NOR as Universal Gate

- NOT A: (A+A)' = A'
- OR A+B: ((A+B)')' = A+B
- AND AB: (A'+B')' = AB [De Morgan]

Why Universal Gates Matter

- Any logic function can be built using ONLY NAND (or ONLY NOR) gates
- Reduces IC fabrication cost — only one gate type needed on chip

Active-High vs Active-Low Signals

- Active-high: signal asserted when HIGH (positive logic, L=0 H=1)
- Active-low: signal asserted when LOW (negative logic, L=1 H=0)

XOR and XNOR Gates

XOR — Exclusive OR

- F = A⊕B = A'B + AB' (output 1 when inputs DIFFER)
- POS form: (A+B)(A'+B')

XNOR — Exclusive NOR (Equivalence)

- F = A⊙B = AB + A'B' (output 1 when inputs are SAME)
- F = (A⊕B)'

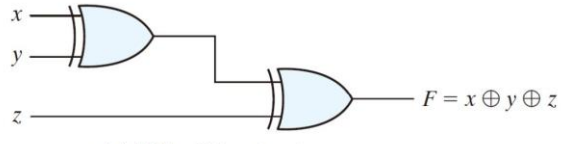
Useful XOR Properties (from Ch.2 Eq.2.25–2.31)

- A ⊕ A = 0 A ⊕ A' = 1
- A ⊕ 0 = A A ⊕ 1 = A'
- A ⊕ B = B ⊕ A (commutative)
- A ⊕ (B⊕C) = (A⊕B) ⊕ C (associative)

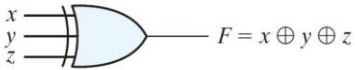
Application

- XOR: adders, parity generators/checkers, comparators
- XNOR: equality detectors, bit comparators

- XOR(XNOR) is an odd(even) function: it is equal to 1 if the inp
have an odd(even) number of 1's



(a) Using 2-input gates



(b) 3-input gate

x	y
0	0
0	0
0	1
0	1
1	0
1	0
1	1
1	1

(c) T

FIGURE 2.8
Three-input exclusive-OR gate

Logic Diagrams from Boolean Expressions

Two-Level AND-OR (SOP) Implementation

- Step 1 — Identify all literals and complements needed
- Step 2 — One AND gate per product term
- Step 3 — One OR gate combines all product terms

Example: F = xy' + x'z

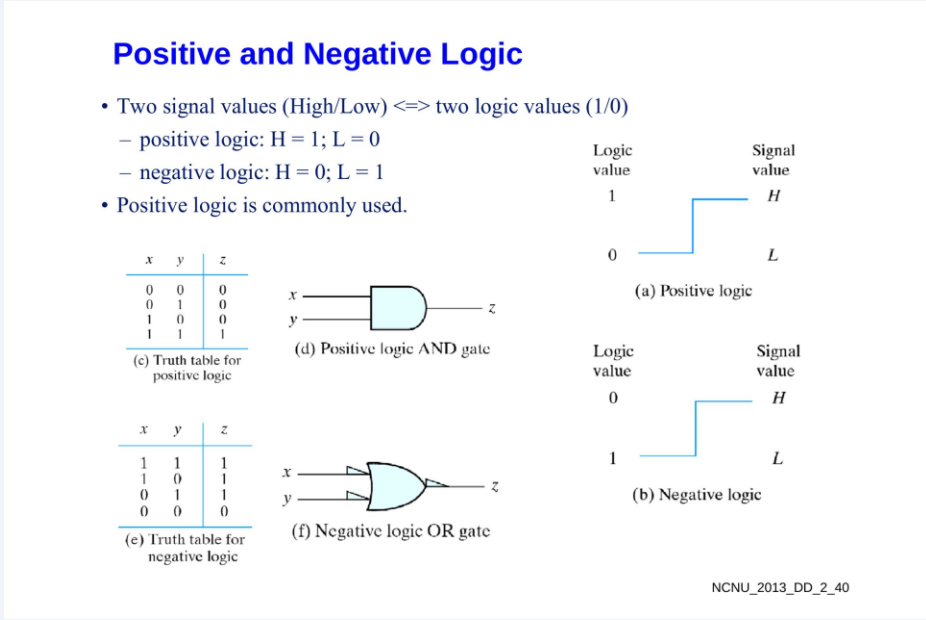
- Gate 1 (NOT): generate y' Gate 2 (NOT): generate x'
- Gate 3 (AND): x · y' → xy'
- Gate 4 (AND): x' · z → x'z
- Gate 5 (OR): xy' + x'z → F

Two-Level OR-AND (POS) Implementation

- One OR gate per sum term, then AND all together

Gate Delay

- SOP / POS = exactly 2 gate levels (AND then OR, or OR then AND)
- Total delay = 2 × per-gate propagation delay



Section 4

Algebraic Manipulation & Simplification

Expression Simplification — Strategy

Goal

- Reduce number of literals and terms → fewer gates, lower cost, faster circuit

Key Techniques

- 1. Combining (Adjacency): AB + AB' = A
- 2. Absorption: A + AB = A
- 3. Simplification: A + A'B = A + B
- 4. De Morgan: convert between sum/product forms
- 5. Consensus: remove redundant terms

Operator Precedence

- Highest: NOT (complement) → then AND → then OR
- A+B·C means A+(B·C) — AND binds tighter than OR

General Approach

- Factor common literals, group adjacently differing terms, then apply absorption

Algebraic Manipulation — Example 2.1 (Mano Ch. 2)

1. x(x' + y) = xx' + xy = 0 + xy = xy

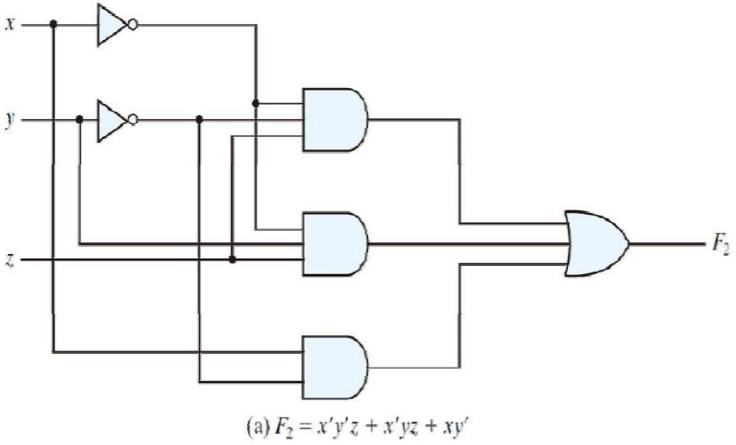
- 2. x + x'y = (x + x')(x + y) = 1(x + y) = x + y
- 3. (x + y)(x + y') = x + xy' + xy + yy' = x(1+y'+y) = x
  - 4. xy + x'z + yz = xy + x'z + yz(x+x') = xy + x'z + xyz + x'yz

= xy(1+z) + x'z(1+y) = xy + x'z

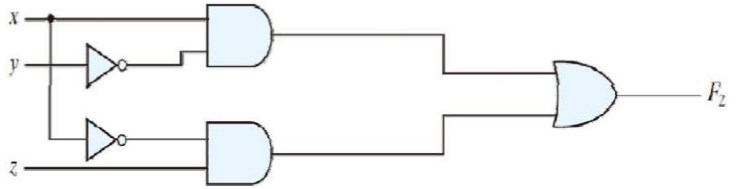
- 5. (x+y)(x'+z)(y+z) = (x+y)(x'+z) [by duality from ex. 4]
- F2 = x'y'z + x'yz + xy' = x'z(y'+y) + xy' = x'z + xy'
- (3 terms, 8 literals) → simplified to 2 terms, 4 literals

Complement of a Function (Examples 2.2 & 2.3)

- F1 = x'yz' + x'y'z → F1' = (x+y'+z)(x+y+z')
- F2 = x(y'z' + yz) → F2' = x' + (y+z)(y'+z')



(a) $F_2 = x'y'z + x'yz + xy'$



(b) $F_2 = xy' + x'z$

Simplification — Worked Examples (Part 2)

Example 4 (De Morgan application)

- (a + bc)' = a'(bc)' = a'(b'+c') = a'b' + a'c'
- Note: (a+bc)' ≠ a'b'+c' (common mistake!)

Example 5 (Nested De Morgan)

- (a(b+z(x+a')))' = a' + (b+z(x+a'))'
  - = a' + b'(z(x+a'))'
  - = a' + b'(z' + (x+a')')
  - = a' + b'(z' + x'a)
  - = a' + b'(z' + x') [T5(a): x'a → x' since a'+x'a=a'+x']

Example 6 (Consensus removal)

- AB + A'CD + BCD = AB + A'CD [T9(a): BCD is consensus]

Example 7: F = (A+B)(A'+C)(B+C)

- (A+B)(A'+C)(B+C) = (A+B)(A'+C) [consensus T9(b)]

Shannon's Expansion Theorem

Theorem 10 — Shannon's Expansion

- $f(x_1,x_2,\ldots,x_n) = x_1 \cdot f(1,x_2,\ldots,x_n) + x_1' \cdot f(0,x_2,\ldots,x_n)$ [cofactor expansion]
- $f(x_1,x_2,\ldots,x_n) = [x_1 + f(0,x_2,\ldots,x_n)] \cdot [x_1' + f(1,x_2,\ldots,x_n)]$

Example: Expand f(A,B,C) = AB + AC' + A'C

- $f = A \cdot f(1,B,C) + A' \cdot f(0,B,C)$
  - $= A(B + C') + A'(C)$
- Expand again on B:
  - $= B[A+A'C] + B'[AC'+A'C]$
  - $= AB + A'BC + AB'C' + A'B'C$

Alternative: Add Missing Literals (Theorem 6)

- $AB = AB(C+C') = ABC + ABC' = m_7 + m_6$
- $AC' = AC'(B+B') = ABC' + AB'C' = m_6 + m_4$
- $A'C = A'C(B+B') = A'BC + A'B'C = m_3 + m_1$
- Therefore $f(A,B,C) = \Sigma m(1,3,4,6,7)$

Section 5

Minterms & Maxterms

Minterms — Definition & Indexing

Definition

- A minterm is a product (AND) term where each of the n variables appears exactly ONCE (complemented or uncomplemented)
- For n variables: exactly 2^n minterms, indexed 0 through 2^n-1

Index Rule for Minterms

- '1' in the binary index → variable appears uncomplemented
- '0' in the binary index → variable appears complemented

Two-Variable Minterms (A, B)

- m_0 = A'B' (00) m_1 = A'B (01)
- m_2 = AB' (10) m_3 = AB (11)

Key Property

- Each minterm equals 1 for exactly ONE input combination
- The sum of ALL minterms = 1 for any set of variable values

Three-Variable Minterms & Maxterms Table

Index	x	y	z
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1

Minterms and Maxterms

- A minterm (standard product): an AND term consists o
  normal form or in their complement form
- For example, two binary variables x and y, has 4 minte
  - $xy, xy', x'y, x'y'$
- n variables can be combined to form 2^n minterms (m_j, j
- A maxterm (standard sum): an OR term; 2^n maxterms (
- Each maxterm is the complement of its corresponding

			Minterms		
x	y	z	Term	Designation	Ter
0	0	0	$x'y'z'$	m_0	$x + y$
0	0	1	$x'y'z$	m_1	$x + y$
0	1	0	$x'yz'$	m_2	$x + y'$
0	1	1	$x'yz$	m_3	$x + y'$
1	0	0	$xy'z'$	m_4	$x' + y$
1	0	1	$xy'z$	m_5	$x' + y$
1	1	0	xyz'	m_6	$x' + y'$

Three-Variable Minterms & Maxterms Table

Minterms and Maxterms

- A minterm (standard product): an AND term consists of all literals in their normal form or in their complement form
- For example, two binary variables x and y, has 4 minterms
  - xy, xy', $x'y$, $x'y'$
- n variables can be combined to form 2^n minterms ($m_j, j = 0 \sim 2^n\text{-}1$)
- A maxterm (standard sum): an OR term; 2^n maxterms ($M_j, j = 0 \sim 2^n\text{-}1$)
- Each maxterm is the complement of its corresponding minterm, and vice versa.

			Minterms		Maxterms	
x	y	z	Term	Designation	Term	Designation
0	0	0	$x'y'z'$	m_0	$x + y + z$	M_0
0	0	1	$x'y'z$	m_1	$x + y + z'$	M_1
0	1	0	$x'yz'$	m_2	$x + y' + z$	M_2
0	1	1	$x'yz$	m_3	$x + y' + z'$	M_3
1	0	0	$xy'z'$	m_4	$x' + y + z$	M_4
1	0	1	$xy'z$	m_5	$x' + y + z'$	M_5
1	1	0	xyz'	m_6	$x' + y' + z$	M_6
1	1	1	xyz	m_7	$x' + y' + z'$	M_7

)D_2

Maxterms — Definition & Indexing

Definition

- A maxterm is a sum (OR) term where each of the n variables appears exactly ONCE (complemented or uncomplemented)
- Dual of minterm: uses + instead of ·

Index Rule for Maxterms (OPPOSITE of minterms)

- '0' in the binary index → variable appears uncomplemented
- '1' in the binary index → variable appears complemented

Two-Variable Maxterms (A, B)

- M_0 = A+B (00) M_1 = A+B' (01)
- M_2 = A'+B (10) M_3 = A'+B' (11)

Key Property

- Each maxterm equals 0 for exactly ONE input combination
- The product of ALL maxterms = 0

Fundamental Relationship

- $(m_i)' = M_i$ and $(M_i)' = m_i$ for the same index i

Purpose of the Index

Index encodes the input combination

- Treat the variable combination as a binary number
- The index tells you the single row where that minterm/maxterm is special

Minterm m_i

- Equals 1 ONLY when inputs match the binary representation of i
- All other 2^n-1 combinations give $m_i = 0$

Maxterm M_i

- Equals 0 ONLY when inputs match the binary representation of i
- All other 2^n-1 combinations give $M_i = 1$

Example — Three Variables

- m_5 = xy'z : equals 1 only when x=1,y=0,z=1 (binary 101 = 5) ✓
- M_5 = x'+y+z': equals 0 only when x=1,y=0,z=1 ✓

Summary

- There are 2^n minterms AND 2^n maxterms for n-variable functions

Section 6

Canonical Forms — SOM & POM

Sum-of-Minterms (SOM) Canonical Form

Definition

- Express F as an OR of minterms for all rows where F = 1
- Also called: Canonical SOP or minterm expansion

Notation

- $F(x,y,z) = \Sigma m(2,3,5,7)$

Derivation — Example 2.4 (Mano): F = A + B'C

- $F = A(B+B') + B'C = AB + AB' + B'C$
  - $= A'B'C + AB'C' + AB'C + ABC' + ABC$
  - $F(A,B,C) = \Sigma(1, 4, 5, 6, 7) = m_1 + m_4 + m_5 + m_6 + m_7$

Textbook Functions (Table 2.4)

- $f_1(x,y,z) = x'y'z + xy'z' + xyz = m_1 + m_4 + m_7 = \Sigma m(1,4,7)$

Key Property

- Every Boolean function has a UNIQUE canonical SOM
- Focus only on the '1' entries of the truth table

Product-of-Maxterms (POM) Canonical Form

Definition

- Express F as an AND of maxterms for all rows where F = 0
- Also called: Canonical POS or maxterm expansion

Notation

- $F(x,y,z) = \Pi M(0,2,6)$

Derivation — Example 2.5 (Mano): F = xy + x'z

- $F = (xy+x')(xy+z) = (x+x')(y+x')(x+z)(y+z)$
  - $= (x'+y)(x+z)(y+z)$ — expand each sum to include all vars
  - $F(x,y,z) = \Pi M(0,2,4,5) = M_0 \cdot M_2 \cdot M_4 \cdot M_5$

Textbook Functions (Table 2.4)

- $f_2(x,y,z) = x'yz + xy'z + xyz' + xyz = m_3+m_5+m_6+m_7 = \Sigma m(3,5,6,7)$

Key Property

- Every Boolean function has a UNIQUE canonical POM
- Focus only on the '0' entries of the truth table

SOM / POM Truth Table Side-by-Side

x	y	z	f	Minterm (for F=1)	Maxterm (for F=0)
0	0	0	0	—	$M_0 = x+y+z$
0	0	1	0	—	$M_1 = x+y+z'$
0	1	0	1	$m_2 = x'yz'$	—
0	1	1	1	$m_3 = x'yz$	—
1	0	0	0	—	$M_4 = x'+y+z$
1	0	1	1	$m_5 = xy'z$	—
1	1	0	0	—	$M_6 = x'+y'+z$
1	1	1	1	$m_7 = xyz$	—

Conversions Between Canonical Forms

SOM → POM

- Minterm indices where F=1 ↔ Maxterm indices where F=0
- The two sets are complementary — they partition $\{0,1,\ldots,2^n-1\}$

Rule

- If $F = \Sigma m(i_1,i_2,\ldots)$ then $F = \Pi M$(all remaining indices)

Example (from Ch.2 Eq.2.10)

- $f_1(A,B,C) = \Sigma m(2,3,6,7) = \Pi M(0,1,4,5)$
- $f_2(A,B,C) = \Pi M(0,1,4,5) = \Sigma m(2,3,6,7)$ — same function!

Function Complement

- If $F = \Sigma m(S)$ then $F' = \Pi M(S)$ — same indices, different symbol
- Example: $F = \Sigma m(2,3,6,7) \rightarrow F' = \Pi M(2,3,6,7)$
- Equivalently: $F' = \Sigma m(0,1,4,5)$

Generalisation

- To convert: interchange $\Sigma \leftrightarrow \Pi$ and list the MISSING indices

Truth Table for $F = xy + x'z$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Minterms

Maxterms

Deriving Canonical Forms Algebraically

Method — Expand Using Theorem 6 (ab+ab'=a)

- Multiply each product term by (X+X') for every missing variable X

Example: f(A,B,C) = AB + AC' + A'C → canonical SOP

- AB = AB(C+C') = ABC + ABC' = $m_7 + m_6$
- AC' = AC'(B+B') = ABC' + AB'C' = $m_6 + m_4$
- A'C = A'C(B+B') = A'BC + A'B'C = $m_3 + m_1$
- f = Σm(1,3,4,6,7) (remove duplicate m_6)

Example: f(A,B,C) = A(A+C') → canonical POS

- A = (A+B)(A+B') then expand each: (A+B'+C')(A+B'+C)(A+B+C')(A+B+C) = $M_3M_2M_1M_0$
- (A+C') = (A+B'+C')(A+B+C') = M_3M_1
- f = ΠM(0,1,2,3)

Don't-Care Conditions

- Some input combinations may never occur → don't-care minterms d_i
- f(A,B,C) = Σm(0,3,7) + d(4,5) = M(1,2,6) · D(4,5)

Section 7

Standard SOP & POS Forms

Standard SOP and POS — Definitions

Standard SOP (Sum of Products)

- OR of AND terms; each term may have FEWER than n variables
- More general than canonical SOP — not every term needs to be a minterm
  - Example: F = AB + C (standard SOP, 2 terms)
  - Canonical SOP of same function: F = ABC+ABC'+AB'C+A'BC+... (more terms)

Standard POS (Product of Sums)

- AND of OR terms; each factor may have fewer than n variables
  - Example: F = (A+C)(A'+B) (standard POS)

Canonical vs Standard

- Canonical: unique, each term has ALL n variables
- Standard: not unique, terms may have fewer — result of simplification
- Standard forms use fewer gates (the whole point of simplification!)

Converting Standard → Canonical

- Expand: for any term T missing variable X, write T = TX + TX'

Two-Level Gate Implementations

AND-OR Network (implements SOP)

- Level 1: AND gates (one per product term)
- Level 2: single OR gate
- Fast — exactly 2 gate delays from input to output

OR-AND Network (implements POS)

- Level 1: OR gates (one per sum term)
- Level 2: single AND gate

NAND-NAND Network (equivalent to AND-OR)

- Replace AND-OR by NAND-NAND using De Morgan: F = ((AB)'(CD)')' = AB+CD
- Preferred in practice — NAND gates are cheapest to fabricate

NOR-NOR Network (equivalent to OR-AND)

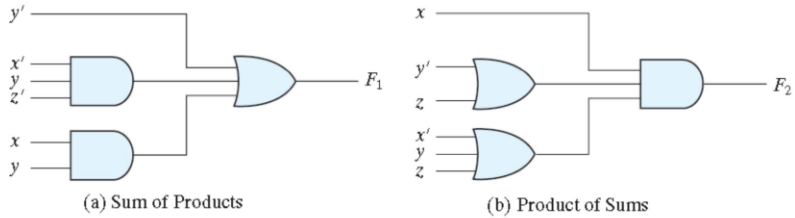
- Replace OR-AND by NOR-NOR using De Morgan

Implementation Procedure for NAND

- 1. Write function as SOM 2. Write minterms 3. Simplify to SOP
- 4. Transform to NAND form 5. Draw NAND diagram

– sum of products $F_1 = y' + xy + x'yz'$

– product of sums $F_2 = x(y' + z)(x' + y + z')$



(a) Sum of Products (b) Product of Sums

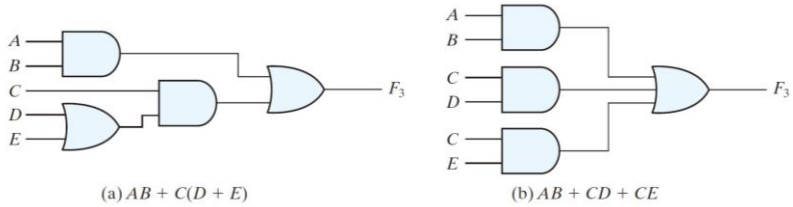
- This circuit configuration is referred to as a *two-level* implementation.

NAND Network — Worked Example

Example (from Ch.2): f(X,Y,Z) = Σm(0,3,4,5,7)

- Step 1 & 2 — write minterms:
  - $f = m_0+m_3+m_4+m_5+m_7 = X'Y'Z'+X'YZ+XY'Z'+XY'Z+XYZ$
- Step 3 — simplify using Theorem 6:
  - $XY'Z' + XY'Z = XY'$; $X'Y'Z' + XY'Z' = Y'Z'$
  - $X'YZ + XYZ = YZ$
  - $f = XY' + Y'Z' + YZ$
- Step 4a — NAND form (using Theorem 4):
  - $f = ((XY')'\cdot(Y'Z')'\cdot(YZ)')'$
- Step 4b — NAND form (using De Morgan / Theorem 8):
  - $f = (XY' + Y'Z' + YZ)'' =$ NAND of NAND of each product term
- Step 5 — draw: three 2-input NAND gates feeding one 3-input NAND gate

- $F_3 = AB + C(D + E)$
  $= AB + C(D + E) = AB + CD + CE$



(a) $AB + C(D + E)$ (b) $AB + CD + CE$

Circuit Synthesis — Design Examples

Burglar Alarm (Ex.2.41)

- A = secret switch closed
- B = safe in normal position
- C = clock between 10:00–14:00
- D = closet door closed

Logic equation

- Alarm = AB' + DC' + DA'
- (safe moved AND secret switch ON)
- (OR closet opened after hours)
- (OR closet opened, switch open)

Smoke Alarm (Ex.2.38)

- S_1, S_2 = smoke detectors (active-low)
- Sprinkler: S_1 NOR S_2
- Fire dept: S_1 NAND S_2

Voter Circuit (Ex.2.42)

- Family vote: hamburger=0, chicken=1
- Majority wins; if Mom & Dad agree, they win
- A=Dad, B=Mom, C=Joe, D=Sue
- F = AB + ACD + BCD

2-bit Adder (Ex.2.43)

- Inputs: $A_1A_o + B_1B_o$
- Outputs: $C_1S_1S_o$
- $S_o = A_o \oplus B_o$
- $S_1 = A_1 \oplus B_1 \oplus (A_oB_o)$
- $C_1 = A_1B_1 + A_1A_oB_o + B_1A_oB_o$

Key insight

- Word description → equations → gates
- Always verify with truth table

Section 8

Analysis of Combinational Circuits

Circuit Analysis Methods

Purpose of Analysis

- Determine the behaviour and verify correctness of an existing circuit
- Convert circuit to a different implementation form

Method 1 — Algebraic Method

- Write a switching expression for each gate output (from inputs forward)
- Combine gate expressions to get the overall output expression
- Simplify using Boolean algebra theorems

Method 2 — Truth Table Method

- Evaluate the output one gate at a time for every input combination
- Fill in the truth table row by row

Method 3 — Timing Diagram Analysis

- Read input waveforms over time
- Propagate through gates level by level
- Verify output waveform matches expected function

Algebraic Analysis — Example (Ex.2.33)

Given circuit for f(a,b,c)

- Internal signal $g_1 = a \oplus b$ (XOR of a and b)
- Internal signal $g_2 = g_1 \cdot c$ (AND)
- Internal signal $g_3 = a \cdot b$ (AND)
- Output: $f = g_2 + g_3$

Simplify

- $f = (a \oplus b)c + ab$
  - $= (a'b+ab')c + ab$ [Eq.2.24]
  - $= a'bc + ab'c + ab$ [distribute]
  - $= a'bc + a(b'c + b)$ [factor a]
  - $= a'bc + a(b+c)$ [T5: b'c+b = b+c]
  - $= a'bc + ab + ac$
  - $= bc + ab + ac$ [T5: a'bc+ac → c(a'+a)b? check]

Result

- $f(a,b,c) = ab + ac + bc$ (majority function!) ✓

Propagation Delay

Definition

- The delay between an input change and the corresponding output change
- Physical characteristic of every real logic gate

Two Key Parameters

- t_PLH = propagation delay, low-to-high output transition
- t_PHL = propagation delay, high-to-low output transition
- Approximation: $t_p \approx (t_{PLH} + t_{PHL}) / 2$

Multi-Level Delay

- Total delay ≈ (number of gate levels) × (per-gate delay)
- Two-level SOP/POS: delay = 2 × t_p (fastest for complex functions)

Logic Families (from Ch.2 Table 2.7)

- TTL (74LS): ~5–10 ns per gate, ~2 mW per gate
- CMOS: ~1–5 ns, extremely low static power
- ECL: fastest (~1 ns), highest power consumption

Fan-in & Fan-out

- Fan-in: max inputs a gate can accept without degraded performance

Algebraic Forms — Complete Summary

Literal

- A variable complemented or uncomplemented: A, A', x, x'

Product Term

- Literals ANDed: AB, AB'C, x'y

Sum Term

- Literals ORed: A+B, A'+B'+C, x+y'

Minterm

- Product term with ALL n variables: AB'C (3 vars)

Maxterm

- Sum term with ALL n variables: A+B'+C (3 vars)

Canonical SOP

- Sum (OR) of minterms only: Σm(indices)

Canonical POS

- Product (AND) of maxterms only: ΠM(indices)

Standard SOP

- OR of product terms (may lack some variables)

Standard POS

- AND of sum terms (may lack some variables)

Conversion Rule

- Σm(S) = ΠM(complement of S)

SOP → POS Algebraically

- Use A+(BC) = (A+B)(A+C)

POS → SOP Algebraically

- Expand (A+B)(C+D) = AC+AD+BC+BD

Don't-Care

- f = Σm(list) + d(don't-care list)
- = ΠM(list) · D(don't-care list)

Complement Identity

- $(m_i)' = M_i \quad (M_i)' = m_i$

Key Takeaways — Boolean Algebra for CSC 231

Boolean Algebra

- 6 postulates define the system
- 10 theorems for simplification
- De Morgan & Consensus are most powerful
- Duality: every law has a dual

Logic Gates

- AND, OR, NOT are fundamental
- NAND and NOR are universal (build anything)
- XOR/XNOR for arithmetic and comparison
- Two-level = 2 gate delays (optimal speed)

Truth Tables

- Unique — one table per function
- Bridge between word description & algebra

Minterms & Maxterms

- $m_i = 1$ for exactly one row; $M_i = 0$ for exactly one row
- $(m_i)' = M_i$ — fundamental duality
- 2^n minterms and maxterms for n variables

Canonical Forms

- SOM: Σm(rows where F=1) — unique
- POM: ΠM(rows where F=0) — unique
- Complementary index sets

Standard Forms & Minimization

- Standard < Canonical in gate count
- Algebraic tools: adjacency, absorption, consensus
- Always verify result matches truth table

Slide 50 — Logic Gate Symbols & IEEE Standard Notation

F=A·B

AB	F
0,0	0
0,1	0
1,0	0
1,1	1

F=A+B

AB	F
0,0	0
0,1	1
1,0	1
1,1	1

F=A'

A	F
0	1
1	0

F=(AB)'

AB	F
0,0	1
0,1	1
1,0	1
1,1	0

F=(A+B)'

AB	F
0,0	1
0,1	0
1,0	0
1,1	0

F=A⊕B

AB	F
0,0	0
0,1	1
1,0	1
1,1	0

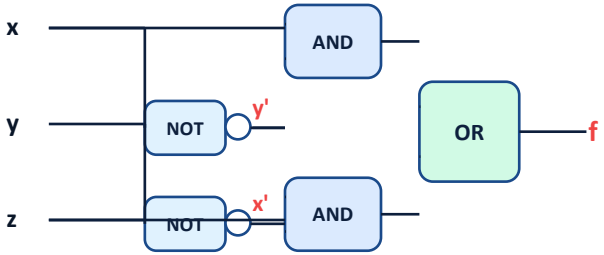
F=(A⊕B)'

AB	F
0,0	1
0,1	0
1,0	0
1,1	1

Note: Circle (○) on output = inversion (NOT) | AND: flat back · OR: curved back · XOR: extra curved line

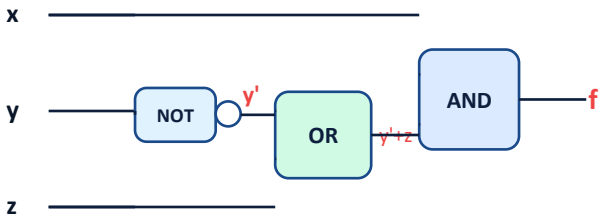
Slide 51 — Logic Diagrams: f = xy'+x'z and f = x(y'+z)

Circuit 1: f = xy' + x'z (SOP — AND-OR, 2 levels)



f = xy' + x'z
- 2 NOT + 2 AND + 1 OR = 5 gates
- Two gate-delay levels (AND → OR)

Circuit 2: f = x(y'+z) (factored — 3 levels)

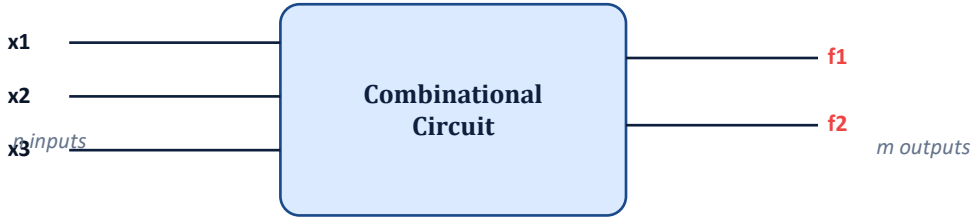


f = x(y'+z) [factored SOP]
- 1 NOT + 1 OR + 1 AND = 3 gates
- Three gate-delay levels; fewer gates

Key Insight: Both circuits implement the same function f = xy'+x'z = x(y'+z). SOP (Circuit 1) has 2 gate delays but uses 5 gates. Factored form (Circuit 2) uses 3 gates but introduces 3 levels of delay.

Slide 52 — Combinational Circuits & Two-Level Implementations

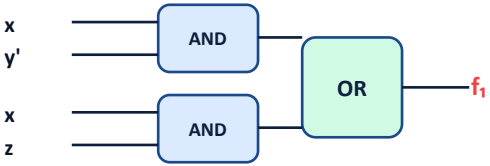
Combinational Circuit Model



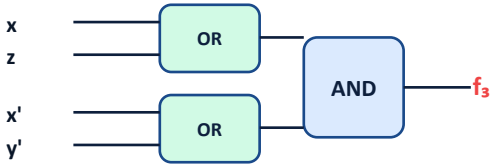
- n inputs, m outputs
- Each output = function of present inputs only
- No feedback / memory (combinational = memoryless)

Two-Level Gate Implementations

$f_1 = xy' + xz$ **(AND-OR / SOP)**



$f_3 = (x+z)(x'+y')$ **(OR-AND / POS)**



AND-OR (SOP): AND gates in Level 1, single OR in Level 2. Implements any SOP expression directly. **OR-AND (POS):** OR gates in Level 1, single AND in Level 2. Implements any POS expression directly.

NAND-NAND ≡ AND-OR | NOR-NOR ≡ OR-AND (both use De Morgan) | Always exactly 2 gate-delay levels

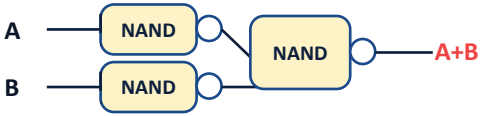
Slide 53 — NAND & NOR Universal Gates: Building AND, OR, NOT

NAND-based Constructions (using De Morgan: (AB)' = A'+B')

NOT from NAND: A NAND A = A'

AND from NAND: ((AB)')' = AB

OR from NAND: A' NAND B' = A+B

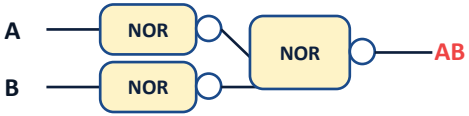


NOR-based Constructions (using De Morgan: (A+B)' = A'B')

NOT from NOR: A NOR A = A'

OR from NOR: ((A+B)')' = A+B

AND from NOR: A' NOR B' = AB



Why Universal Gates? Any Boolean function can be implemented using ONLY NAND gates, or ONLY NOR gates.
This reduces IC fabrication cost — manufacturers need to produce only ONE type of gate.
NAND: preferred for TTL/CMOS SOP circuits · NOR: preferred for CMOS POS circuits · NAND-NAND = AND-OR · NOR-NOR = OR-AND

Slide 54 — SOM & POM: Full Worked Example

Function: f(x,y,z) from truth table

x	y	z	f	Minterm / Maxterm
0	0	0	0	$M_0 = x+y+z$
0	0	1	0	$M_1 = x+y+z'$
0	1	0	1	$m_2 = x'yz'$
0	1	1	1	$m_3 = x'yz$
1	0	0	0	$M_4 = x'+y+z$
1	0	1	1	$m_5 = xy'z$
1	1	0	0	$M_6 = x'+y'+z$
1	1	1	1	$m_7 = xyz$

F=1 → contributes to SOM

F=0 → contributes to POM

Canonical Forms

Sum-of-Minterms (SOM) = Σm(2,3,5,7)

f = x'yz' + x'yz + xy'z + xyz

= Σm(2,3,5,7)

Product-of-Maxterms (POM) = ΠM(0,1,4,6)

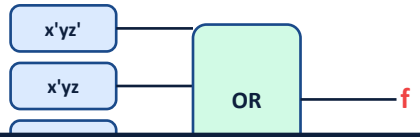
f = (x+y+z)(x+y+z')(x'+y+z)(x'+y'+z)

= ΠM(0,1,4,6)

Conversion Identity: Σm(2,3,5,7) = ΠM(0,1,4,6)

Complement: f' = Σm(0,1,4,6) = ΠM(2,3,5,7) | $(m_i)' = M_i$

Circuit: f = x'yz' + x'yz + xy'z + xyz (4-input AND-OR)



Steps: (1) Truth table → identify F=1 rows → (2) write minterm for each → (3) OR all minterms = canonical SOM

Homework #2

Problems from Mano — Digital Design, Chapter 2

- 2.2 (e) — Prove the Boolean identity using postulates
- 2.2 (f) — Prove the Boolean identity using postulates

Boolean Theorems & Simplification

- 2.4 (d) — Simplify the Boolean expression to minimum literals
- 2.4 (e) — Simplify the Boolean expression to minimum literals

Canonical Forms & Logic Gates

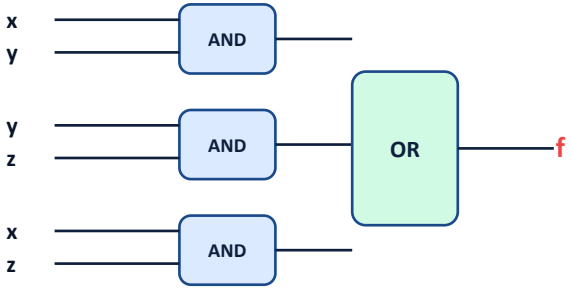
- 2.9 (c) — Express function in canonical SOM and POM form
- 2.11 (b) — Implement the Boolean function with logic gates

Standard Forms, XOR & ICs

- 2.14 (b), 2.14 (c), 2.22 (b), 2.28 — Standard forms, XOR properties

Slide 55 — Circuit Analysis & 2-Level vs 3-Level Implementation

Algebraic Circuit Analysis: Derive f from gates



Step-by-step:

$g_1 = xy$

$g_2 = yz$

$g_3 = xz$

$f = g_1+g_2+g_3$

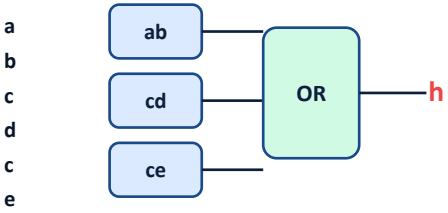
$f = xy+yz+xz$

Result: f(x,y,z) = xy + yz + xz (Majority function — F=1 when ≥2 inputs are 1)

Canonical SOM: Σm(3,5,6,7) | Canonical POM: ΠM(0,1,2,4)

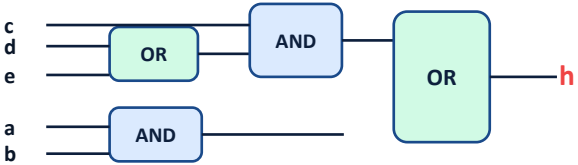
2-Level vs 3-Level: h = ab+cd+ce

2-Level (SOP): h = ab + cd + ce



3 AND + 1 OR | 2 gate levels | 6 literals

3-Level (factored): h = ab + c(d+e)



1 OR + 2 AND + 1 OR | 3 levels | 5 literals — fewer gates!

Trade-off: 2-Level SOP is faster (2 gate delays) but uses more gates (more chip area). Factored 3-Level uses fewer gates (lower cost) but is slower (3 delays). *Fan-in constraints (max inputs per gate) also force multi-level designs on large circuits.*

Thank You

Questions & Discussion — CSC 231 Boolean Algebra